

Monitoring, Metrics & Alerts

Read the signal correctly • Explain the outage exercise • Know current coverage

SAY IT IN THE INTERVIEW

"I exposed health and Prometheus metrics through Actuator. Prometheus scrapes Inventory Service every five seconds, and Grafana uses those metrics for monitoring and service-down alerts. I exercised the alert behavior locally. The current Compose configuration keeps SMTP disabled unless I explicitly configure it."

01 WHO DOES WHAT?

- **Actuator** exposes operational endpoints. **Micrometer** provides application/JVM meters. **Prometheus** collects time-series samples. **Grafana** queries and visualizes them, and evaluates configured alerts.
- Both services expose **/actuator/health** and **/actuator/prometheus**. The committed scrape configuration targets only **Inventory :8082** every **5s**.

02 THE AVAILABILITY SIGNAL

- **up=1**: the latest target scrape succeeded. **up=0**: it failed. A failed scrape can mean the app, network or metrics endpoint is unavailable.
- Example query: **up{job="inventory-service"}**. Use the same job label when tracing the alert condition.
- A service-stop exercise should show failed scrapes, a firing alert after its configured evaluation/pending period, then recovery after successful scrapes resume. Five-second scraping is not a guaranteed five-second notification.

03 WHAT I CAN SHOW FROM THIS PROJECT

- Committed captures include Prometheus target-up, a Grafana dashboard folder and separate Firing/Normal alert states. They support the local monitoring walkthrough.
- The dashboard image shows the folder listing, not panel graphs. Separate alert-rule identifiers are not proof of one correlated firing-to-recovery history.
- Grafana UI configuration persists in **grafana-data**. Dashboard/alert provisioning files are not committed; a fresh machine needs that setup.

04 HOW I ANSWER AN OPERATIONS FOLLOW-UP

- "I check health, scrape status and logs first, then trace the affected event through the databases and Kafka flow. A healthy process can still have a stuck business workflow."
- **Future additions**: Order-service scraping, outbox backlog/age, DLT rate and consumer-lag monitoring. These are useful next signals, not claimed current dashboards.
- Alert evaluation and email delivery are separate. **GF_SMTP_ENABLED=false** is the default; an email contact-point label does not prove message delivery.

YAAD RAKHO: process health, scrape health and business completion are different.